

One-Page Brief: Slack Constraints & Novel Predictions

This sheet distills the analysis of the foliation+axion TOE framework into immediate opportunities for novel predictions. We identify the **least constrained directions** in parameter space (from elasticity and PCA scans) and show how they map to potential observable signatures.

Least Constrained Gates (Elasticity Scan)

- Photon LIV spurion (ζ_F / M_F^2): birefringence/dispersion still has headroom.
- Fermion LIV spurion (ϵ / M_ψ): sidereal modulation of atomic clocks/comagnetometers still possible.
- Axion-like coupling $g_{ag\gamma}$ (for $\theta_4 \approx \text{axion}$): BM-2 sits just below current CAST/helioscope bounds.

PCA of Survivor Cloud

- PC1: mixed photon LIV – 0.3 fermion LIV: softest combination.
 - PC2: c_2 vs c_4 wedge: survives provided $|c_{2,4}| \leq 1e-8$.
- These directions are prime for designing observables where signals add coherently.

Edge Projection (Benchmarks)

- BM-1 (spectator): pushing photon LIV to 95% bound → birefringence $\sim 1e-15$ rad over GRB baselines.
- BM-2 (axion-like): at $g_{ag\gamma} \approx 5e-12 \text{ GeV}^{-1}$, predicts polarization rotation of order $1e-3$ – $1e-2$ rad in X-ray polarimetry.
- BM-3 (stressed): small but nonzero daily modulation within reach of next-gen atomic clocks.

Targets & Tactics

- Photon LIV: GRB/FRB polarimetry & high-energy dispersion over Gpc baselines.
- Fermion LIV: daily modulation searches in ultra-precise comagnetometers and optical lattice clocks.
- Axion-like: X-ray polarimetry (IXPE/eXTP), helioscope upgrades (IAXO), haloscopes probing $g_{ag\gamma}$ just below $1e-11 \text{ GeV}^{-1}$.